Project Title:

Open Source Audit — Linux Kernel

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Course: Open Source Software

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1. Introduction

The **Linux Kernel** is the core part of the Linux operating system. It controls how hardware and software interact with each other. Every application in a Linux system depends on the kernel to function properly.

This project is focused on studying the Linux Kernel as an open-source project. It includes both theoretical understanding and practical implementation through shell scripts.

Part A — Origin and Philosophy

A1. Problem the Software Was Created to Solve

Before Linux existed, operating systems like Unix were expensive and restricted. Students and developers could not easily access or modify them. This created a limitation for learning and innovation.

The Linux Kernel was created by Linus Torvalds in 1991. He wanted a free and open alternative to Unix that anyone could use and improve.

The main problems it solved:

- High cost of proprietary operating systems
- No access to source code
- Limited collaboration among developers

Linux allowed users to freely study and modify the system, which encouraged global collaboration.

A2. The License — What It Actually Says

The Linux Kernel is licensed under **GNU General Public License Version 2 (GPLv2)**.

This license provides four important freedoms:

1. Freedom to run the program
2. Freedom to study how it works
3. Freedom to modify the code
4. Freedom to share copies

A company cannot take Linux, modify it, and sell it without sharing the source code. This is because GPL requires all modified versions to remain open.

GPL vs MIT License:

- GPL → requires sharing of modifications
- MIT → allows private modifications

Free vs Freedom:

- Free software does not only mean no cost
- It also means users have control and rights over the software

A3. Ethics of Open Source

Open source is based on sharing knowledge and collaboration. It allows developers worldwide to contribute and improve software.

However, there are debates:

- Not all software can be open source
- Companies may use open-source software to build profitable products

For example, companies use Linux to build commercial systems, which raises questions about fairness.

Open source supports innovation because developers can build on existing work instead of starting from zero.

Part B — Linux Footprint

The Linux Kernel is deeply integrated into the system.

- **Installation:** Comes pre-installed with Linux distributions
- **Directories:**
  - /boot → contains kernel files
  - /lib/modules → contains kernel modules
- **User:** Runs with root privileges
- **Service:** Loaded during system boot
- **Update model:** Updates are provided through package managers like apt

This shows how the Linux Kernel operates at the system level and controls core functionality.

Part C — The FOSS Ecosystem

The Linux Kernel is part of a large open-source ecosystem.

- **Dependencies:**
  - GCC compiler
  - GNU tools
- **Used in:**
  - Servers
  - Android devices
  - Cloud systems

It is also part of the web ecosystem and supports many applications.

The development of Linux is supported by a global community of developers who contribute through version control systems.

Part D — Open Source vs Proprietary

Features	Linux Kernel	Proprietary OS
Cost	Free	Paid
Security	Transparent	Closed
Support	Community-Based	Company-Based
Modification	Allowed	Restricted
Control	User Control	Company Control

Final Verdict-The Linux Kernel is highly flexible and secure. It is suitable for real-world use and encourages learning and innovation.

Shell Scripts Section

Script-1:

This script displays system information such as kernel version, user, uptime, and distribution. It uses variables and command substitution.

```bash
#!/bin/bash
# Script 1: System Identity Report
# Author: Your Name | Course: Open Source Software

STUDENT_NAME="Rishabh Singh"
SOFTWARE_CHOICE="Linux Kernel"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DISTRO=$(grep PRETTY_NAME /etc/os-release | cut -d '"' -f2)
DATE=$(date)

echo "================================"
echo " Open Source Audit — $Rishabh Singh"
echo "================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Kernel   : $KERNEL"
echo "Distro   : $DISTRO"
echo "User     : $USER_NAME"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "License  : GNU General Public License v2 (GPLv2)"
```

Script-2:

This script checks if the Linux Kernel is installed and displays its details. It uses conditional statements and case structure.

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector

PACKAGE="linux-image-$(uname -r)"

if dpkg -l | grep -q "$PACKAGE"; then
    echo "Linux Kernel package is installed."
    dpkg -l | grep "$PACKAGE"
else
    echo "Kernel package info not found via dpkg."
fi

echo "Kernel Version:"
uname -r

echo "Kernel Details:"
uname -a

case "linux" in
    linux) echo "Linux Kernel: the foundation of all Linux systems" ;;
    apache) echo "Apache: web server powering the internet" ;;
    mysql) echo "MySQL: open-source database system" ;;
    git) echo "Git: version control built for developers" ;;
    *) echo "Unknown package" ;;
esac
```

Script-3:

This script analyzes system directories and displays their size and permissions. It uses loops and file handling commands.

```bash
#!/bin/bash
# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/src" "/boot")

echo "Directory Audit Report"
echo "----------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist"
    fi
done

# Kernel-related directory check
if [ -d "/boot" ]; then
    echo "Kernel files in /boot:"
    ls -l /boot | grep vmlinuz
fi
```

Script-4:

This script reads log files and counts occurrences of specific keywords. It helps in monitoring system logs.

```bash
#!/bin/bash
# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "Error: File not found."
    exit 1
fi

while read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -5
```

Script-5:

This script collects user input and generates a personalized open-source statement saved in a file.

```bash
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer the questions:"

read -p "1. Tool you use daily: " TOOL
read -p "2. Meaning of freedom: " FREEDOM
read -p "3. What will you build: " BUILD

DATE=$(date)
OUTPUT="manifesto_$(whoami).txt"

echo "On $DATE, I believe in open source." > $OUTPUT
echo "I use $TOOL daily, which shows the power of community-driven software." >> $OUTPUT
echo "For me, freedom means $FREEDOM." >> $OUTPUT
echo "I aim to build $BUILD and share it freely with others." >> $OUTPUT

echo "Manifesto saved to $OUTPUT"
cat $OUTPUT
```

Conclusion

This project provided a deeper understanding of the Linux Kernel and open-source principles. It also helped in learning shell scripting and system-level operations.

The Linux Kernel plays a major role in modern computing and continues to evolve through community contributions.

References

- GNU Project
- Linux Documentation
- Open Source Initiative